Phase A – AWS IAM Credentials and Identity Context

Objective: Review how the AWS CLI interacts with IAM credentials, understand user-based and role-based authentication, verify the current AWS identity context, and compare static and temporary credentials.

1. AWS CLI and IAM Credentials

The AWS CLI requires valid credentials to authenticate requests to AWS services. Credentials may be provided through environment variables, AWS CLI profiles, credential files, AWS IAM Identity Center (SSO), or IAM roles. The AWS CLI uses the available credential configuration to determine which identity is used for a command.

2. User-Based vs. Role-Based Authentication

Aspect	IAM User	IAM Role
Identity	Persistent AWS identity	Identity that can be assumed by an authorized user, service, or workload
Credentials	Can use long-term access keys	Typically provides temporary security credentials
Common Use	Individual users and some legacy workloads	AWS services, workloads, SSO, and cross-account access
Expiration	Credentials remain valid until rotated, disabled, or deleted	Temporary credentials expire after a defined period

3. AWS STS Identity Verification

The following command was executed to confirm the identity associated with the current AWS CLI session:

aws sts get-caller-identity

Command output (sensitive account information may be redacted when sharing publicly):


```
{
  "UserId": "AIDA5VAX2ZKE04TQBOAAO",
  "Account": "938500344456",
```

"Arn": "arn:aws:iam::938500344456:user/test-user"
}

4. Identity Context

The returned ARN contains the pattern arn:aws:iam::ACCOUNT_ID:user/USERNAME. This confirms that the current AWS CLI session is authenticated as an IAM user, rather than an assumed IAM role.

Current IAM identity: test-user

Identity type: IAM User

5. Static vs. Temporary Credentials

Static (long-term) credentials consist of an Access Key ID and Secret Access Key that remain valid until they are rotated, disabled, or deleted. They are commonly associated with IAM users and require appropriate security controls to prevent exposure.

Temporary credentials are issued for a limited period and generally consist of an Access Key ID, Secret Access Key, and Session Token. They are commonly provided through AWS Security Token Service (STS), such as when an IAM role is assumed. Temporary credentials are generally preferred for temporary access and workloads because they reduce the risks associated with long-lived credentials.

6. Completion Checklist

☑ Executed aws sts get-caller-identity

☑ Identified current IAM user/role: IAM user test-user

☑ Summarized differences between static and temporary credentials

Conclusion

Phase A was completed successfully. The AWS CLI identity was verified using AWS STS, and the current identity was confirmed as the IAM user test-user. The differences between long-term static credentials and short-lived temporary credentials were also documented.